SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

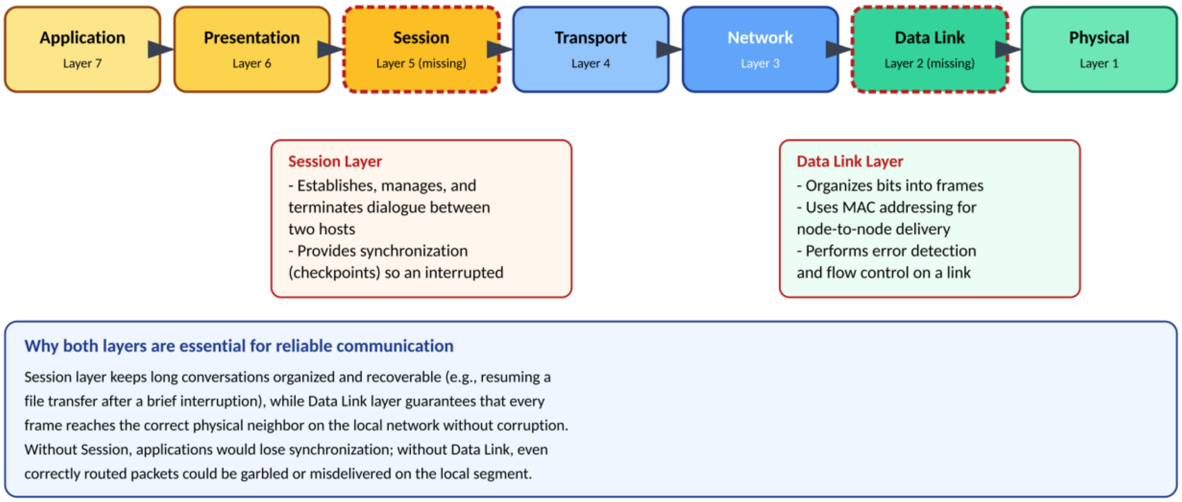
I **KGODDAVULA NITISH KANTH /192512232**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.MAHESH

Annexure A

1. Missing Layers in the OSI Concept Map

The given map — Application → Presentation → ______ → Transport → Network → ______ → Physical — is missing the Session layer (between Presentation and Transport) and the Data Link layer (between Network and Physical). The completed map and each layer's essential function are shown below.



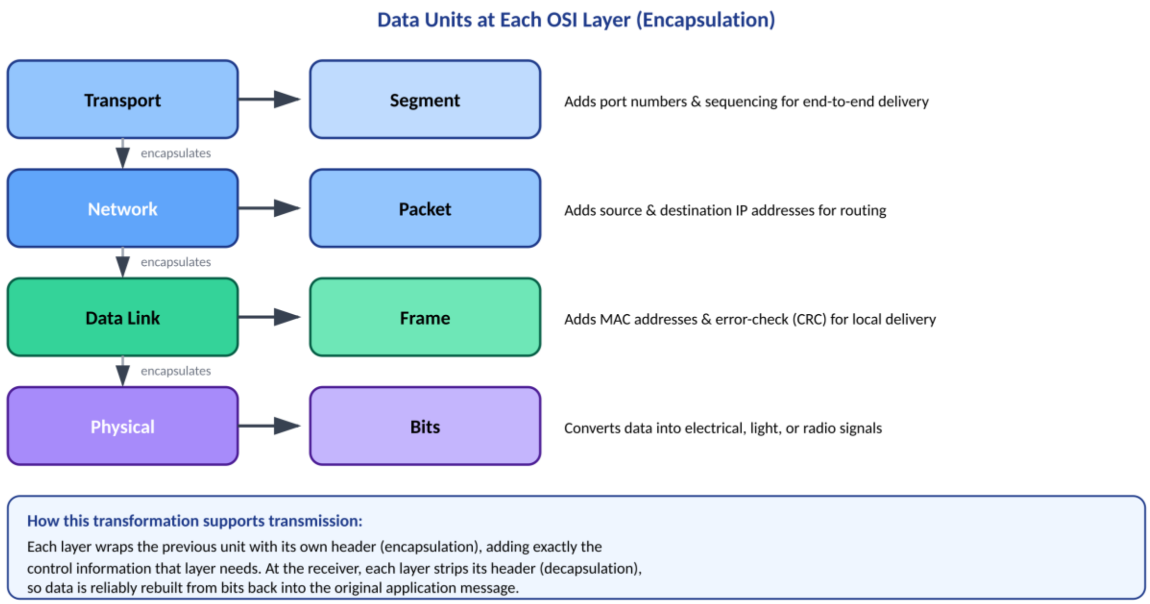
Analysis

- Session Layer: Opens, manages, and cleanly closes a dialogue between two communicating applications. It inserts checkpoints so that if a long transfer is interrupted, communication can resume from the last checkpoint instead of restarting from zero.
- Data Link Layer: Packages network-layer packets into frames, attaches physical (MAC) addresses for the next hop, and performs error detection (e.g., CRC) plus flow control across a single physical link.

Together the two layers close different reliability gaps: Session protects the logical continuity of a conversation, while Data Link protects the physical accuracy of each hop. Without Session, applications would lose synchronization during interruptions; without Data Link, even correctly routed packets could be corrupted or misdelivered on the local segment.

2. Concept Map: Data Units Across OSI Layers

Each layer of the OSI model works with its own protocol data unit (PDU). As data descends the stack, every layer wraps the unit received from the layer above with its own header — a process called encapsulation.



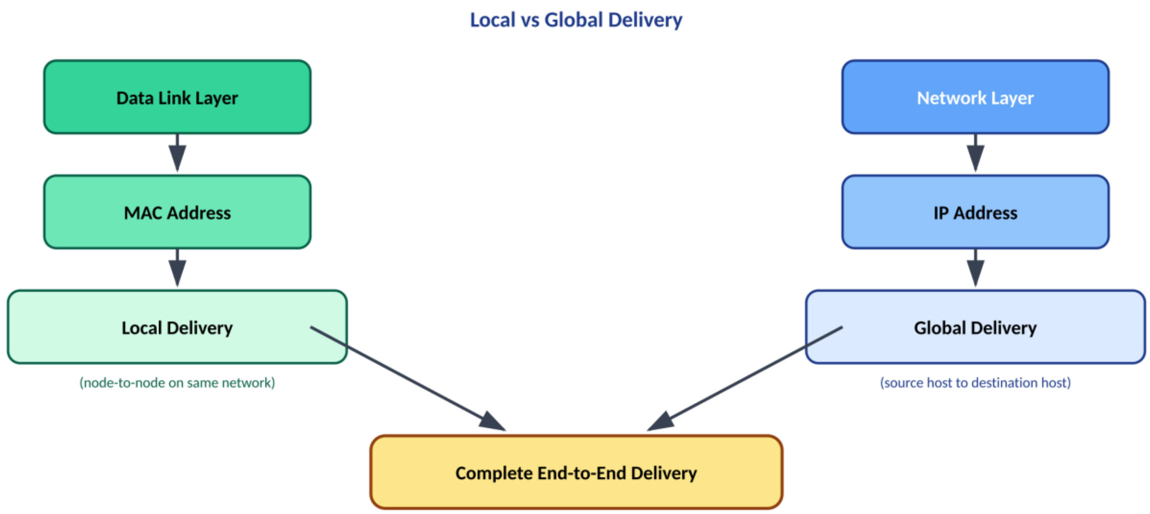
Analysis

- Segment (Transport Layer): Adds port numbers and sequence numbers so the correct application process and byte order can be identified at the destination.
- Packet (Network Layer): Adds source and destination IP addresses so routers can forward the unit across multiple networks toward the destination host.
- Frame (Data Link Layer): Adds MAC addresses and an error-checking trailer so the unit can move correctly across one physical link.
- Bits (Physical Layer): Converts the frame into a raw stream of electrical, optical, or radio signals for transmission on the medium.

This layered transformation — encapsulation going down the stack at the sender and decapsulation going up the stack at the receiver — lets each layer solve one problem independently (addressing, routing, error checking, signalling) while still cooperating to deliver the original message intact.

3. Concept Map: Local vs Global Delivery

The given map splits addressing into two parallel chains: Data Link → MAC Address → Local Delivery, and Network → IP Address → Global Delivery.



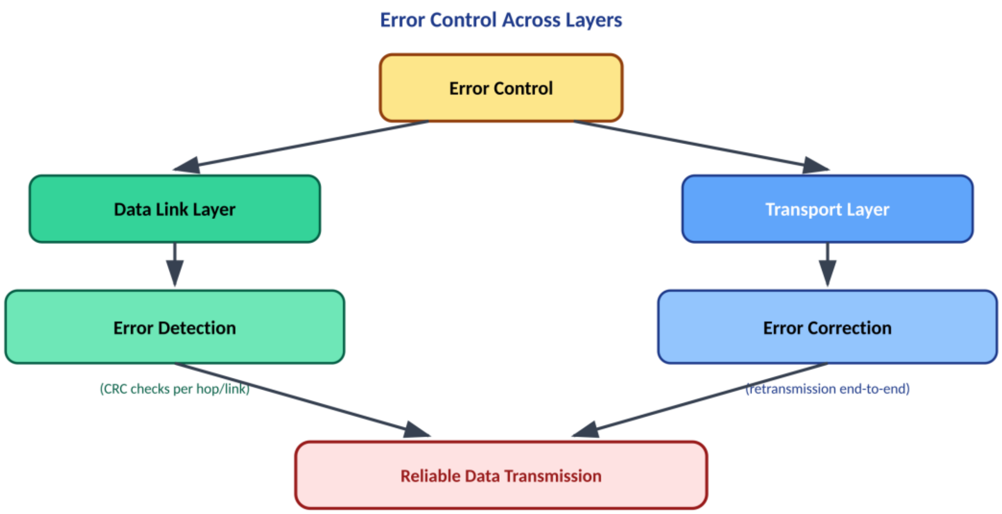
Analysis

- MAC addresses are physical, hardware-burned addresses used only within a single local network (LAN). They let the Data Link layer deliver a frame to the correct device on the same physical segment.
- IP addresses are logical addresses assigned per network. They let the Network layer route a packet across multiple interconnected networks toward a distant host, regardless of the physical medium in between.

Complete, end-to-end delivery needs both: IP addressing gets the packet to the correct destination network, while MAC addressing (re-applied at every hop along the route) makes the final local delivery to the correct device on the destination's LAN. Removing either address type breaks delivery — global routing without MAC addressing could not reach a specific device on the destination network, and MAC addressing alone could never route beyond the local segment.

4. Concept Map: Error Control Across Layers

The map shows Error Control splitting into two branches: Data Link Layer → Error Detection, and Transport Layer → Error Correction.



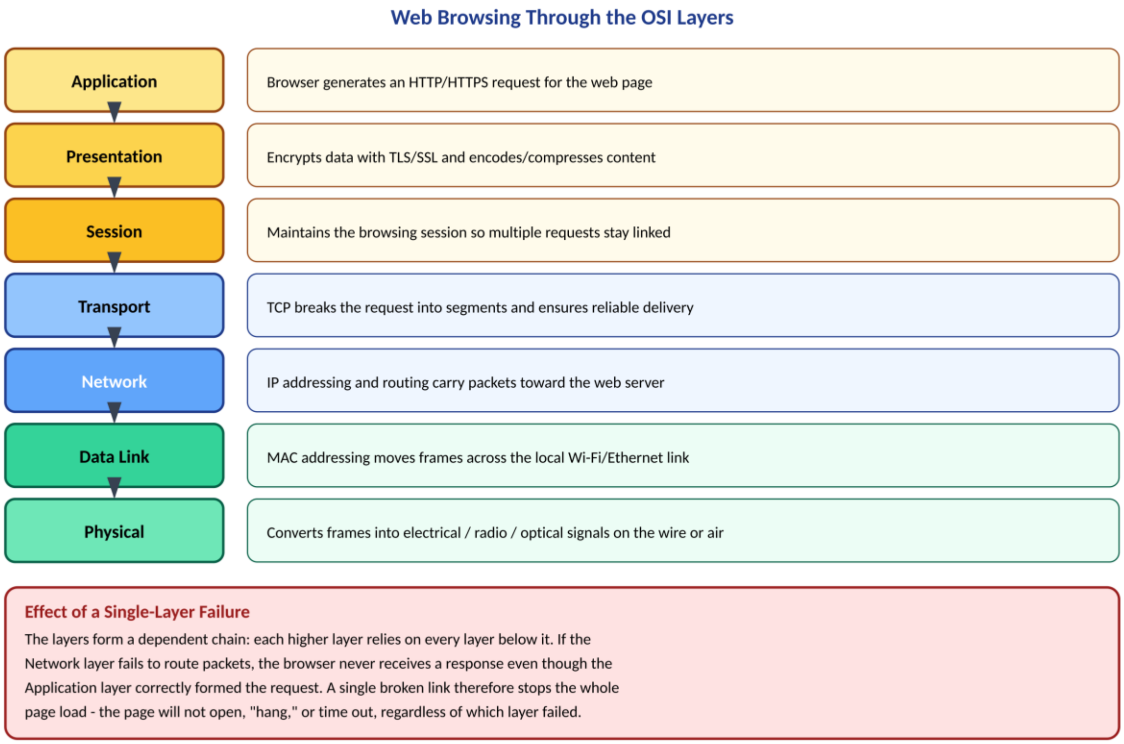
Analysis

- Data Link Layer (Error Detection): Uses checksums such as CRC on every individual link (hop) to detect whether a frame was corrupted in transit. It typically only flags or discards a bad frame rather than fixing it.
- Transport Layer (Error Correction): Works end-to-end between the original sender and final receiver. Using acknowledgements, sequence numbers, and retransmission (e.g., in TCP), it corrects errors — including loss — that slip past the local checks.

The two layers form a two-stage defense: Data Link filters out damage on each individual link early and cheaply, while Transport guarantees that whatever reaches the destination application is complete and correct, even if some link along the path failed to catch an error. This division of labour keeps error handling efficient (local, fast checks) without sacrificing end-to-end reliability (global correction).

5. Concept Map: Web Browsing Through the OSI Layers

Loading a web page exercises every OSI layer, from generating the HTTP request down to sending signals over the network medium.



Analysis

Because each layer depends on the one below it, the layers form a strict chain rather than independent parallel paths. If any single layer fails — for example, the Network layer cannot route the packet, or the Physical layer loses signal on the cable — every layer above it is starved of usable data, even if it performed its own job correctly.

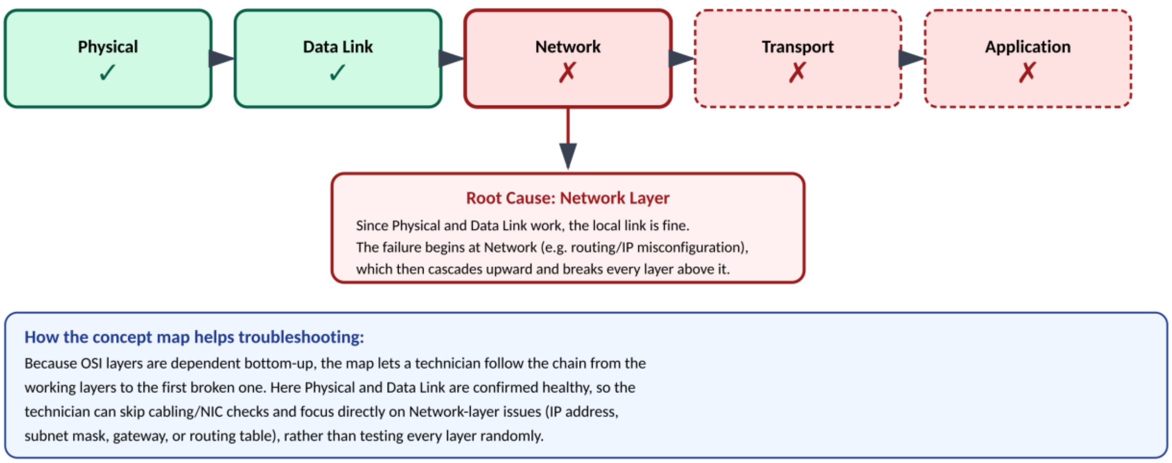
- A Physical-layer fault (unplugged cable, dead Wi-Fi radio) prevents any signal from leaving the device at all.
- A Network-layer fault (misconfigured IP, unreachable route) means the correctly-formed HTTP request never reaches the server.
- A Presentation-layer fault (failed TLS handshake) can block the page from loading securely even though lower layers work perfectly.

In every case the visible symptom is the same to the user — the page fails to load — which is why systematic, layer-by-layer analysis (rather than guessing) is essential to find the actual point of failure.

6. Concept Map: Troubleshooting a Network Issue

The map records diagnostic results as the request moves up the stack: Physical ✓→ Data Link ✓→ Network ✗→ Transport ✗→ Application ✗

Troubleshooting with the Layer Concept Map



Analysis

Since OSI layers depend on the ones below them, a failure at one layer will always cause every layer above it to fail as well, even if those upper layers are functioning correctly in isolation. Reading the map from the bottom up:

- Physical (✓) and Data Link (✓) confirm the local cabling, network card, and immediate link partner are all working — the signal and frames are moving correctly on this segment.
- Network (✗) is the first layer to fail, pointing to an IP-level problem: a misconfigured address, subnet mask, default gateway, or a routing failure.
- Transport (✗) and Application (✗) fail only because they never receive usable data from the broken Network layer — they are symptoms, not separate root causes.

This is the practical value of a layered concept map for troubleshooting: instead of testing everything at once, a technician can start at the lowest layer and move upward only until they find the first failure. That failure point is almost always the true root cause, letting scarce troubleshooting time be spent on the Network layer configuration rather than re-checking cabling or application settings that are already known to be fine.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation